Rendering SSSOM ontology mappings as RDF named graphs and RDF 1.2 triple terms

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Introduction

The Simple Standard for Sharing Ontological Mappings (SSSOM) (Matentzoglu et al., 2022) is widely used in the life sciences to share mappings between terms of different datasets and ontologies. A SSSOM file is a hybrid artefact: a YAML metadata header (carried in #-prefixed comment lines) followed by a tab-separated (TSV) table of mappings. Although the standard carries ontological commitments, the files are most often consumed as plain dataframes.

SSSOM already defines an RDF/OWL representation that describes a *mapping set as a whole*. During the GOBLIN project at the BioHackathon, we explored two alternative, complementary RDF renderings that instead make the individual *mapping* the first-class object, and we compared the design trade-offs between them. This report documents that exploration. **It is an early, first-draft proposal intended to start a conversation rather than a finished specification** (see Discussion).

Approach

The native SSSOM format

The core of each TSV row — `subject_id`, `predicate_id`, `object_id` — *is* an RDF triple; the remaining columns (`mapping_justification`, labels, . . .) and the header (provenance, confidence, the `curie_map`) are metadata about that triple or about the set. A mapping is therefore, ontologically, *a statement about a statement*, which can be expressed in RDF at two different layers.

Two renderings

We built a single streaming serializer that emits both representations from the SSSOM TSV files. **For transparency: the serializer was generated by prompting a large language model (Claude); it is a starting point to be audited, not a reference implementation.** It parses the YAML header, expands every CURIE to a full IRI using the file's authoritative `curie_map`, and writes RDF.

Named graphs (TriG). Each mapping set becomes one named graph (graph IRI = `mapping_set_id`); the core triples live inside the graph and set-level provenance is asserted about the graph IRI in the default graph. This sits at the RDF *dataset* layer and faithfully models `sssom:MappingSet`. It is universally supported (plain SPARQL 1.1 `GRAPH`), but per-mapping metadata is not attached.

RDF 1.2 (Turtle-star). Each mapping triple is asserted and annotated with an RDF 1.2 triple term (Hartig et al., 2021), using the `{| ... |}` annotation syntax that desugars to a reifier (`_:m rdf:reifies <<( s p o )>>`) carrying the justification, labels and a back-link to the set. This sits at the *abstract-syntax* layer, models `sssom:Mapping`, and is lossless — at the cost of a ~5x statement blow-up and a dependency on RDF-star-aware tooling.

CURIE expansion and IRI-unsafe local names

The header CURIE map is treated as authoritative. However, some local-name parts are not valid inside an IRI — the canonical case is the `SMILES:` "prefix", which encodes chemical structures rather than identifiers and contains characters (`/`, `#`, `[`, `]`, `=`, ...) that IRI syntax forbids. We percent-encode the local part (per RFC 3987), leaving the namespace literal, so the result is a syntactically valid IRI.

Sources

We used two sources. The EBI Ontology Lookup Service (OLS) (Côté et al., 2006), which publishes per-ontology SSSOM extracts, was processed in full. We also began extracting ontology cross-references from Wikidata as SSSOM; this was not completed within the hackathon and is reported as future work.

Results

The full set of OLS SSSOM extracts — **271 mapping sets, 6,238,000 mappings** — was rendered into both formats in ~91 s. Both outputs were validated by parsing with a strict RDF 1.2 parser (pyoxigraph). The renderings, schema diagrams (drawn in ShEx and, for RDF 1.2, in ShEx *style* since ShEx 2.1 has no triple-term construct yet), and tooling notes are published in the project code repository (https://github.com/andrawaag/GOBLIN_Oviedo_hackathon_2026); the gzipped RDF is distributed there via Git LFS. The two renderings are loaded into Oxigraph (Pellissier Tanon, 2024), which serves a SPARQL endpoint, and queried from a browser with Comunica (Taelman et al., 2018), which can additionally federate the local endpoint with remote endpoints such as Wikidata.

Table 1: Summary of the two RDF renderings of the OLS SSSOM extracts.

Property	Named graphs (TriG)	RDF 1.2 (Turtle-star)
RDF layer	dataset / context (quads)	abstract syntax (triple terms)
Granularity	one graph per mapping set	per mapping (`{| ... |}`)
Models	`sssom:MappingSet`	`sssom:Mapping`
Metadata carried	set-level provenance	justification + labels (lossless)
Tooling	universal (SPARQL 1.1)	needs RDF-star engine
Size	1.5 GB (~6.24M quads)	3.7 GB (~34M statements)

Discussion

The two renderings are not competing encodings of the same idea: they attach metadata at different layers of the RDF stack and model different SSSOM entities (named graph <-> `MappingSet`, triple term <-> `Mapping`). They therefore *compose* — the named-graph file is the portable, partition-by-source view, and the RDF 1.2 file is the lossless, per-mapping master.

Relative to the existing SSSOM/RDF representation, our approach makes two deliberate scope choices. First, we do not support mappings between literals (e.g. `"hello"@en skos:closeMatch "hallo"@nl`), which require generalized RDF that RDF 1.2 triple terms do not provide; as most datasets and tooling do not use generalized RDF, we accept that this excludes a small number of files. Second, we describe each mapping independently rather than describing the file as a whole.

This work is explicitly a **first-draft proposal** and the outputs require validation before reuse:

- **IRIs must be minted.** The serializer emits two placeholder IRIs under `w3id.org` that we do not own: a convenience predicate `https://w3id.org/sssom/mapping_set` (not a standard SSSOM slot) and a `https://w3id.org/sssom/.well-known/curie/...` fallback for unresolved CURIEs. The reused `sssom:` slot IRIs are assumed to match the official SSSOM model and must be verified.
- **Modelling is not yet validated.** Parsing with an RDF 1.2 parser confirms syntactic validity, not correct modelling; the shapes need review against the SSSOM community.
- **Propagatable slots are not yet implemented.** Header slots such as `license` and `mapping_date`, and `extension_definitions`, are part of the proposed approach but not yet emitted by the script.

Future work includes completing the Wikidata-to-SSSOM extraction and federating it with the OLS mappings through Comunica, aligning the vocabulary with the official SSSOM/RDF model, minting the required IRIs, and validating the RDF 1.2 shapes once ShEx (or an equivalent) gains triple-term support.

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References

Côté, R. G., Jones, P., Apweiler, R., & Hermjakob, H. (2006). The ontology lookup service, a lightweight cross-platform tool for controlled vocabulary queries. *BMC Bioinformatics*, *7*, 97. https://doi.org/10.1186/1471-2105-7-97 **[cito:usesDataFrom]**

Hartig, O., Champin, P.-A., Kellogg, G., & Seaborne, A. (2021). *RDF-star and SPARQL-star*. W3C Community Group Final Report. https://www.w3.org/2021/12/rdf-star.html **[cito:citesAsAuthority]**

Matentzoglu, N., Balhoff, J. P., Bello, S. M., Bizon, C., Brush, M., Callahan, T. J., Chute, C. G., Duncan, W. D., Evelo, C. T., Gabriel, D.others. (2022). A simple standard for sharing ontological mappings (SSSOM). *Database*, *2022*, baac035. https://doi.org/10.1093/database/baac035 **[cito:citesAsAuthority]**

Pellissier Tanon, T. (2024). *Oxigraph: A SPARQL database and RDF toolkit*. https://github.com/oxigraph/oxigraph. **[cito:usesMethodIn]**

Taelman, R., Van Herwegen, J., Vander Sande, M., & Verborgh, R. (2018). Comunica: A modular SPARQL query engine for the web. *The Semantic Web – ISWC 2018*, 239–255. https://doi.org/10.1007/978-3-030-00668-6_15 **[cito:usesMethodIn]**

Author contributions

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